WRRA-AGAMA v0.3

Numerical closure and a first Gaia sample comparison

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Scope: actual stellar distribution-function and Poisson iteration, followed by public Gaia comparisons. Numerical convergence is demonstrated at two resolutions. Full joint star-count and velocity validation is **not yet established**.

Assessment stage	What was actually calculated
Validation inputs	Original Galactic-disk masses and radial scales; literature finite heights; explicit, fixed trial stellar DFs. A direct Gaia DR3 sample is used for a subsequent comparison.
WRRA transformation	At each baryonic state, apply the prescribed WRRA residual response; reconstruct its spherical effective density and potential; update the live stellar f(J) and repeat.
Outputs	Converged forces and local densities at two resolutions; orbit and Jeans diagnostics; actual selected Gaia star counts and vertical velocities.
Falsification conditions	Non-positive source, non-convergence, grid sensitivity, invariant violation, or disagreement with properly matched observational tracers and selection.

Fixed physical specification; improved numerical controls

The scale a_T = 1.191812669 x 10^-10 m/s^2 and the original constitutive response are retained. Gas and bulge are static; two stellar QuasiIsothermal DFs are live. Initial stellar heights are 0.3/0.9 kpc; gas sech-squared scale parameters are 0.085/0.045 kpc. The trial radial-dispersion amplitudes are 35/55 km/s at 8.2 kpc. These DF choices are explicit model inputs, not new measurements or microscopic derivations. The spatial residual closure is spherical.

The Python interface was extended to expose AGAMA's existing internal integration controls; the physical formulae were unchanged. Density integration uses relative tolerance 10^-4 and up to 10^6 evaluations. The stellar potential grid begins at z = 0.08 kpc. Both runs start from the stored v0.2 density and perform five additional iterations, satisfying the 0.2% local criterion twice consecutively. The patch and warm-start inputs are included.

At R = 8.2 kpc	Base grid	Refined grid
Additional warm-start iterations / checks	5 / 16 of 16	5 / 16 of 16
Circular speed [km/s]	215.56532	215.62089
Kz(1.1 kpc)/(2 pi G) [Msun/pc^2]	71.82974	71.83659
True column within |z| = 1.1 kpc [Msun/pc^2]	79.31428	79.31056
Total midplane Poisson density [Msun/pc^3]	0.0790443	0.0790715

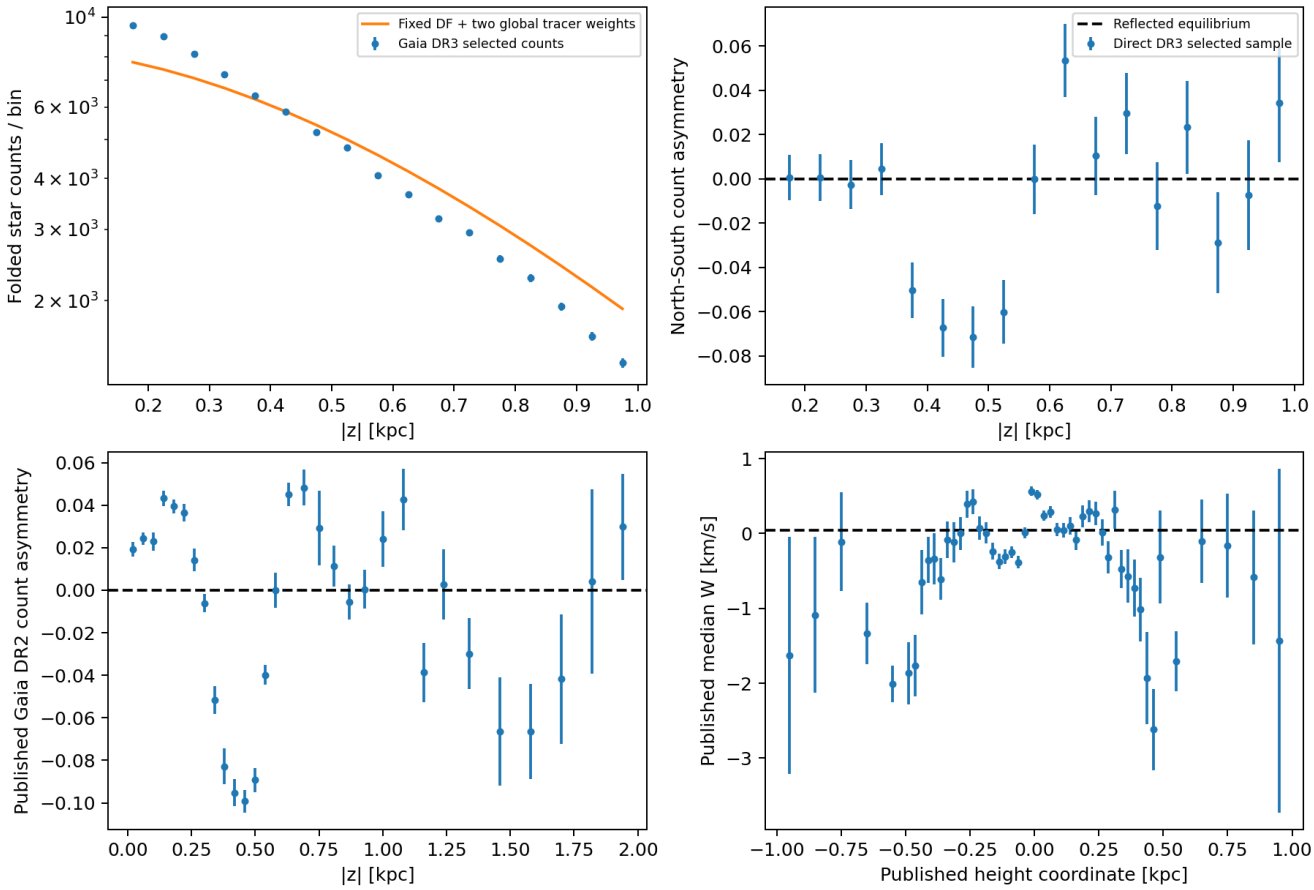
Kz/(2 pi G) is a force-equivalent surface density; it is not the integrated mass column. Their difference retains the radial term of the cylindrical Poisson equation. Displayed digits support reproduction and do not represent observational accuracy.

Numerical checks and actual Gaia data

Across the two grids, the maximum force difference on 20 probes is **0.0515%**; circular speeds differ by at most **0.0268%** over 5-25 kpc. Total midplane density differs by **0.0344%**, resolving the earlier numerical sensitivity. Four test orbits each cover ten nominal periods and pass energy and axial-angular-momentum checks. A sampled direct-moment Jeans residual is at most 2.19%; approximate actions are not exact invariants.

ESA Gaia DR3 sample	Retrieved / compared
Explicit archive selection	Cylinder radius 0.25 kpc; G = 7-17; BP-RP = 1.0-1.4; uncorrected absolute G = 5-6; parallax S/N above 10; RUWE below 1.4.
Retrieved stars / with radial velocities	90,113 / 72,603
RV uncertainty below 3 km/s	48,794 stars
Count comparison	79,716 stars in 17 paired height bins, |z| = 0.15-1.0 kpc. Two global tracer weights fitted; no gravity refit or per-bin adjustment.

Actual Gaia comparisons: equilibrium background and remaining perturbations



Upper panels: directly retrieved Gaia DR3 counts and raw sample asymmetry. The geometry-only fixed-DF count template does not fit the selected counts; its Poisson deviance is 1747.2 after two global weights. Lower panels: independent author-released Gaia DR2 perturbation tables from Bennett & Bovy. These comparisons use different selections and are not combined into a joint likelihood.

Interpretation and review boundary

Claim	Supported status
Actual potential-density/AGAMA implementation	Yes. The prescribed WRRA response, density inversion, stellar DF and Poisson iteration are executed.
Density convergence at both tested resolutions	Yes, on the stated probes and numerical tests. No continuum or global theorem is claimed.
An unchanged frozen field during iteration	No. The field evolves; the response law and DF parameters remain fixed.
Zero added physical assumptions	No. Finite thickness, trial DFs, spherical closure and global boundary regularization are explicitly adopted.
Star counts and vertical velocities fully validated	No. The selected count profile is not reproduced by the current template, full selection/population mapping is open, and vertical perturbations are absent from this equilibrium background.

What the observational comparison establishes

Mass density is not itself a star count. The current geometry-only conversion fits two global population weights but does not forward-model extinction, luminosity-colour distributions, RUWE cuts or the full Gaia/RVS completeness. Its count-profile discrepancy cannot yet be assigned uniquely to gravity. A population-matched tracer DF and a verified selection function are required.

The reflected equilibrium background predicts zero North-South asymmetry and zero vertical streaming. The published DR2 table contains count asymmetries reaching 9.92%. Diagonal residual sums are 1692.2 for 31 count-asymmetry bins and 437.0 for 50 velocity bins after one global velocity offset. Covariance is not supplied, so no formal joint significance is claimed. The same limitation applies to a conventional reflected equilibrium control. This is a missing perturbation sector, not a unique test of WRRA against standard gravity.

The initial finite-height circular speed is 222.16 km/s; the coupled result is about 215.62 km/s. The live stellar DF changes the initially specified spatial density. The old linearized rotation reference has RMS difference about 5.18 km/s over 8.25-25 kpc; it is not an individual-data likelihood.

Concrete next steps

1. Forward-model the selected stellar populations and Gaia/RVS selection, then calibrate the tracer DF against density and dispersion data.
2. Test a fixed calibrated model on held-out observables.
3. Add and test a physically specified non-equilibrium sector if the goal includes the measured vertical waves. Calibration to verified observations is legitimate; it must remain explicit.

Sources and reproduction

AGAMA source: GalacticDynamics-Oxford/Agama, commit f302756b8af2b763db58e278e30478517dc8eea3, plus the included precision-control interface patch.

WRRA radial model: Wonsik-Choi-janefather/wrra-galactic-disk, commit 2528be62a70c1579308389621e75988f1f777c82. Upstream manuscript: doi.org/10.5281/zenodo.23003875.

Gaia DR3: ESA Gaia Archive, gaiadr3.gaia_source; full ADQL query, raw public-source rows and checksums included. Gaia DR3 overview: doi.org/10.1051/0004-6361/202243940.

Bennett & Bovy: arxiv.org/abs/1809.03507; author repository morganb-phys/VWaves-GaiaDR2, commit 203c2e111c24b99351871e562d4efe193fed586b.

Finite-height model family: McMillan (2017), doi.org/10.1093/mnras/stw2759, and official AGAMA data/McMillan17.ini. The full McMillan model is not copied.